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Global Data-Driven Analysis of Brain Connectivity during Emotion Regulation by EEG Neurofeedback

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Abstract

Background: Emotion regulation by neurofeedback involves interactions among multiple brain regions, including prefrontal cortex and subcortical regions in the limbic system. Previous studies focused on connections of specific brain regions like amygdala with other brain regions.

New method: EEG neurofeedback is used to upregulate positive emotion through induced happiness by retrieving positive autobiographical memories and fMRI data acquired simultaneously. A global data-driven approach, group independent component analysis (ICA), is applied to fMRI data and functional network connectivity is estimated. This study discovers all connections among independent components involved in emotion regulation.

Results: The proposed approach identified all functional networks engaged in positive autobiographical memories and evaluated effects of neurofeedback. The results revealed two pairs of networks with significantly different functional connectivity among emotion regulation blocks (relative to other blocks of experiment) and between experimental and control groups (FDR-corrected for multiple comparisons, q=0.05). Functional network connectivity distribution (FNCD) showed significant connectivity differences between neurofeedback blocks and other blocks, revealing more synchronized brain networks during neurofeedback. During emotion regulation, significant functional connectivity changes were found in and between prefrontal, parietal, temporal, occipital, and limbic networks.

Comparison with existing methods: While the results are consistent with those of previous model based studies, some of the connections found in this study were not found previously. These connections are between a) occipital (fusiform, cuneus, middle occipital, and lingual gyrus) and other regions including limbic system/sub-lobar (thalamus, hippocampus, amygdala, caudate, putamen, insula, and ventral striatum), prefrontal/frontal cortex (DLPFC, VLPFC, and OFC), inferior parietal, middle temporal gyrus and b) PCC and hippocampus.

Conclusions: Using fMRI during EEG neurofeedback, this study provided a global insight to brain connectivity for emotion regulation. The brain networks interactions may be used to develop connectivity-based neurofeedback methods and alternative therapeutic approaches, which may be more effective than the traditional activity-based neurofeedback methods.

Keywords: Emotion regulation, simultaneous EEG-fMRI, neurofeedback, independent component analysis (ICA), functional network connectivity, autobiographical memories.

1- Introduction

Emotion regulation consists of extrinsic and intrinsic complex processes used to modify emotional states or mood which includes maintenance, enhancement, or inhibition of one's behavioral response. Several emotion regulation strategies are proposed in (Gross, 1998) including situation selection and modification, attentional deployment, cognitive change, and response modulation. Based on the previous studies and cognitive models proposed for emotion regulation, emotion regulation involves interactions among different brain regions of prefrontal cortex and subcortical regions, specially limbic system (Li et al., 2016; Linhartová et al., 2019). There are various intervention techniques for emotion regulation including pharmacological therapy, deep brain stimulation (DBS), transcranial magnetic stimulation (TMS), and neurofeedback with different advantages and disadvantages (Papo, 2019).

Neurofeedback is a self-brain training technique used to modulate and enhance brain function through receiving neural activity feedback (Scheinost et al., 2013). Compared to other alternative interventions with side effects, neurofeedback offers various advantages such as non-invasive intervention, long lasting effects, low risk with few side effects and investigating and understanding the relationship between brain function and behavior (Luigjes et al., 2019; Sulzer et al., 2013; Van Doren et al., 2019). Several studies have demonstrated the feasibility and effectiveness of neurofeedback for emotion regulation and treatment of mental disorders (Enriquez-Geppert et al., 2017; Gapen et al., 2016; Hartwell et al., 2016; Herwig et al., 2019; Kinreich et al., 2012; Lackner et al., 2016; Liew et al., 2016; Mennella et al., 2017; Nicholson et al., 2016; Quaedflieg et al., 2015; Sitaram et al., 2017; Young et al., 2018b; Zuberer et al., 2018). The training results can be quantified as changes in the activity/function of the brain networks or behavior in medical (clinical symptoms) or non-medical applications.

EEG and fMRI are two main neuroimaging modalities used in several studies of neurofeedback with demonstrated effects on the brain functions for treatment of mental disorders or emotion regulation (Enriquez-Geppert et al., 2017; Gapen et al., 2016; Linhartová et al., 2019; Morgenroth et al., 2020; Wang et al., 2019). Recent advances in data acquisition has made it possible to perform simultaneous recording of EEG and fMRI especially for neurofeedback (Goldman et al., 2000; Ives et al., 1993). Simultaneous EEG-fMRI utilizes the advantages of both modalities especially high temporal and spatial resolutions (Ritter and Villringer, 2006). It provides the ability to access deep brain regions and design new neurofeedback paradigms which can be used later using EEG alone to avoid the cost of fMRI neurofeedback (Zotev et al., 2014). The ability to modulate and apply interventions on the involved brain circuits through EEG neurofeedback along with simultaneous fMRI, will provide valuable information to understand the interactions among the brain circuits in various clinical applications.

According to several researches and recently fMRI review study, several brain circuits consisting of deep brain regions (including limbic system) and prefrontal regions are involved with the normal and abnormal behaviors as well as the emotion regulation process (Linhartová et al., 2019; Phillips et al., 2008). From the limbic system, amygdala plays a key role in generating and regulating emotional responses beside other deep regions like thalamus and insula. Amygdala interacts with several limbic and prefrontal regions during different emotional stimulus and regulation states (Zotev et al., 2011a). Prefrontal regions especially dorsolateral, ventrolateral, and dorsomedial prefrontal cortex and orbitofrontal cortex with reciprocal connections to amygdala play key roles in different steps of emotion regulation like generation of emotional states (Kohn et al., 2014; Linhartová et al., 2019; Svoboda et al., 2006). Also, in several mental disorders including neurological and psychiatric diseases, abnormality has been reported in various brain networks and connectivity of the brain regions (Yamashita et al., 2017; Broyd et al., 2009; Fornito et al., 2015). Since cognitive functions and behaviors are related to the brain networks, understanding the interactions among the brain networks may be used to develop new therapeutic paradigms to modify specific brain networks (like connectivity based neurofeedback) and also can be used as a biomarker for mental disorders (Kim et al., 2015; Koush et al., 2017; Megumi et al., 2015). Despite existence of several emotion regulation studies using neurofeedback, interactions among brain regions/networks involved in emotion regulation are still unclear (Li et al., 2016; Sulzer et al., 2013). Therefore, comprehensive analysis of fMRI data is needed to have a better understanding of the brain function as a result of neurofeedback.

This study proposes to use a global data-driven approach to identify functional networks and discover all connections among functional networks that are involved in emotion regulation as a result of neurofeedback. To this aim, EEG neurofeedback is used to upregulate positive emotion through induced happiness (by retrieving positive autobiographical memories) while acquiring fMRI data simultaneously. Independent component analysis (ICA) is a data driven method that does not need prior model of data to extract distinct, spatially independent functional networks whose time courses are not necessarily independent. This method reveals functional network connectivity (FNC) as the temporal correlation/dependency among the time courses of independent components (ICs) (Calhoun et al., 2001; Meda et al., 2009). Most of the previous emotion regulation studies used model based methods for functional connectivity analysis and found connections among specific brain regions, like amygdala and rest of the brain (Ferri et al., 2016; Li et al., 2016; Murphy et al., 2016; Seeley et al., 2019; Zotev et al., 2013, 2011a). The aim of this study is to discover the effect of neurofeedback on connectivity of functional networks and compare the FNC between emotion regulation blocks and other blocks of experiment. We hypothesize that emotion regulation using neurofeedback changes the connectivity of distinct functional networks in/between prefrontal and limbic regions. To the best of our knowledge, this is the first study based on ICA to examine FNC and quantify the connectivity changes due to emotion regulation via EEG neurofeedback. Our approach can be outlined as follows: We utilize group ICA to identify spatially independent functional networks. Then, the pair-wise correlation between the time courses of these functional networks among all participants is calculated to discover significant connections as a result of neurofeedback. This is done by comparing FNC of emotion regulation blocks and other blocks of the experiment and also those of the experimental and control groups.

2- Methods and Materials

2.1 Research Protocol and Participants

The research protocol was approved by the ethics committees of the Iran University of Medical Sciences, Tehran, Iran. Eighteen healthy males (age 26.7 ± 3.6 years) as the experimental group and fourteen healthy males (mean age 27 ± 3.8 years) as the control group participated in this study. The exclusion criteria included existence or past history of major psychiatric or neurological disorder, drug or alcohol abuse during the past year, brain surgery, and problems related to being imaged by MRI. All participants were right-handed with normal or corrected-to-normal vision. Also, before the experiment, two psychometric tests including Beck's Depression Inventory (Craven et al., 2005) and General Health Questionnaire - 28 (GHQ-28) (Nazifi et al., 2014) were completed by each participant. The mean ± standard deviation of Beck's Depression Inventory and GHQ-28 were

6.8 ± 3 and 2.2 ± 2, respectively. The participants were normal according to the scores of the Beck's Depression Inventory and non-psychiatric according to the scores of the GHQ-28 test.

Participants in the experimental group experienced emotion regulation with real EEG neurofeedback whereas the control group experienced the same protocol with the sham EEG neurofeedback. The experiment contained 10 runs of three blocks, namely, rest, view, and upregulation. Before the experiment, each participant was asked to write at least ten positive autobiographical memories. Different from similar paradigms used in (Young et al., 2014; Zotev et al., 2016; Zotev et al., 2014), during blocks of view and upregulation, pictures of positive autobiographical memories (expressed during interviews) were presented to increase the effectiveness of retrieving positive autobiographical memories. The duration of the rest, view, and upregulation blocks were 20, 40, and 60 seconds, respectively. During the rest block, the participants were asked to relax without remembering anything. In the view block, two pictures (from those announced during interview) were presented for 40s and the participants were asked to see them without remembering anything. In the upregulation block, two images similar to the view block were presented and participants tried to remember the autobiographical positive memories related to the presented images and increase the height of the neurofeedback bar. Neurofeedback was presented only in the upregulation period based on the approach–withdrawal hypothesis (Davidson, 1998; Davidson et al., 1990) and was calculated as the difference between the EEG power in the right and left hemispheres in the alpha frequency band in 2s time windows, updated every 1s with 50% overlap between the consecutive windows. The paradigm is shown in Fig. 1.

2.2 Data Acquisition

The MRI data were acquired using a 3 Tesla Scanner (Prisma, Siemens, Erlangen, Germany) located in the National Brain Mapping Lab (NBML), Tehran, Iran. Functional MRI were acquired using a T2*-weighted gradient-echo, echo-planar (EPI) pulse sequence (TR = 2000 msec, TE = 30 msec, matrix size = 64× 64×30, and voxel size = 3.8×3.8×4 mm). During the 10 runs of the experiment session, 650 volume images were acquired. Structural images were acquired using a gradient-echo, T1-weighted MPRAGE pulse sequence (TI = 1100 msec, TR = 1810 msec, TE = 3.47 msec, and voxel size = 1×1×1 mm).

The EEG data were recorded simultaneously with fMRI using an MRI-compatible EEG system (Brain Products, München, Germany). The EEG cap had 63 electrodes according to the 10-20 system and one ECG electrode. The impedances of the EEG Electrodes were maintained below 5K Ohms and the EEG signal was recorded at 5K samples/sec. The task was presented by Psychtoolbox program through a coil mounted display.

2.3 Real-Time Data Analysis

Due to practical limitations, neurofeedback was provided only based on the EEG signal. The BrainVision RecView software (Brain Products GmbH) was used to remove MRI and ballistocardiogram artifacts from the 64-channel EEG data in real time using a moving average method. As the mean head displacement obtained in offline analysis was 0.41 $\pm$ 0.17 mm, the result of moving average template subtraction did not differ significantly from the counterpart methods used in offline analysis (Moosmann et al., 2009; Niazy et al., 2005).

The denoised data was down-sampled to 250 samples/sec. Then, the difference in powers of the channels F4 and F3 in the alpha band (frontal asymmetry) were calculated every 1 sec using a 2-sec moving window in the upregulation blocks with respect to the baseline; where the baseline was the average difference between the powers of the channels F4 and F3 in the previous view block. The frontal asymmetry index was presented as a neurofeedback bar during the upregulation blocks.

2.4 Offline Data Analysis

As depicted in Fig. 2, the offline analysis consisted of several steps including fMRI preprocessing, group ICA analysis, IC selection, cross correlation between time courses of ICs for each participant in the experimental group, statistical test to determine the significant functional connectivity between ICs in the upregulation versus view blocks for the experimental group and finally statistical test between the experimental and control groups to reveal the effect of neurofeedback. Pre-processing of the fMRI data was performed in FSL and included slice-timing correction, rigid-body motion correction with 6 degrees of freedom (mean head motion for the experimental group was 0.40 ± 0.13 mm and for the control group was 0.42 ± 0.22 mm), normalization to the Montreal Neurological Institute (MNI 2mm) standard space, spatial smoothing using an 8 mm full-width at half-maximum Gaussian kernel, and temporal high-pass filter (cut-off = 0.005 Hz) (Smith et al., 2004; Fsl, 2006). Following preprocessing, a group ICA method was applied on all subjects to identify spatially independent components (networks) using the GIFT toolbox (Calhoun et al., 2001). This step included data reduction using principle component analysis (PCA) and evaluation of the reliability of independent components (ICs) using the ICASSO software, which investigated reliability of ICA estimates by clustering and visualization of the results of multiple runs of ICA. The number of independent components was estimated to be 51 using the minimum length criteria (Li et al., 2007). Then, back-reconstruction was used to estimate the spatial map and time course of each IC for each subject. Before connectivity analysis, the ICs corresponding to noise, motion artifacts, breath or heartbeat were removed by visual inspection of the group independent components, yielding 43 ICs for further analysis. Next, the cross correlation (FC) between the time courses of ICs for each participant was calculated.

2.5 Estimation of Functional Network Connectivity

Spatial ICA decomposed the fMRI data into spatially independent maps (functional networks) and the corresponding time courses. According to (Jafri et al., 2008), functional network connectivity (FNC) between independent components can be estimated as the Pearson correlation of the IC's time courses. Using this approach, FNC was estimated between each pair of 43 ICs both in the view and upregulation blocks separately for each participant in the experimental group. Then, for each connection (each pair of ICs), a statistical test (paired t-test) was performed to specify if this connection was significantly different between upregulation and view among all participants of the experimental group. The procedure was repeated for 43 × (42-1)/2 = 903 different connections, by considering the FDR-correction for multiple comparisons at q = 0.05.

To reveal the effect of neurofeedback, the extracted significant links (upregulation-view) for the experimental group were compared with those of the control group and connections significantly different between the experimental and control groups were identified as the neurofeedback effect of emotion regulation. For this purpose, a two-sample t-test was performed between the experimental and control groups on the significant connections (obtained by comparing the upregulation and view blocks of the experimental group) and FDR-corrected for multiple comparisons at q = 0.05. Estimation of functional network connectivity between the independent components (Pearson correlations of the IC's time courses) and calculation of statistical tests were done using MATLAB and Statistics Toolbox Release 2016b.

3- Results

The FNC results revealed 32 significant connections among 43×(43-1)/2=903 connections between the upregulation and view blocks out of which two were significantly different between the experimental and control groups.The significant connections were between IC9 and IC22 with p-valueupregulation versus view of experimental group = 1.7×10-3 (p-valueexperimental group versus control group = 3×10-3),and between IC26 and IC44 with p-valueupregulation versus view of experimental group = 8.7×10-5 (p-valueexperimental group versus control group = 2.6×10-4). The spatial maps and the brain regions for these components are presented in Fig. 3 and Table 1.

The connectivity changes between the two pairs of networks (ICs 9-22 and 26-44) for the view and upregulation blocks in the experimental group are presented in Fig. 4. This figure illustrates that as a result of neurofeedback, the connectivity between ICs in the upregulation blocks is higher than the view blocks.

Fig. 5 shows the differences between functional connectivity of upregulation and view for ICs 9-22 and 26-44, in the experimental and control groups. This figure illustrates that the real neurofeedback has increased the differences between functional connectivity of upregulation and view of ICs more than sham neurofeedback.

To have a better understanding of the distinction between functional connectivity of upregulation and view blocks, the functional network connectivity distribution (FNCD) of two identified connections (between ICs 9-22 and 26-44) for the experimental group is presented in Fig. 6. For this purpose, the range of correlation coefficients for the view and upregulation blocks are divided into 15 sections (bins), separately. Then, the number of correlation coefficients in each bin is plotted and a Gaussian curve fitted to the resulting FNCD of the view and upregulation blocks. This figure shows that the FNCD in the upregulation blocks is higher and distributed towards positive connectivity than the view blocks, which means that brain networks are more synchronized during the upregulation (neurofeedback) blocks.

4- Discussion

This study used group ICA as a framework to determine spatial maps of the brain functional networks and evaluate the interactions of the brain networks in two conditions of emotion regulation. In contrast to previous model-based studies (Young et al., 2014; Zotev et al., 2016, 2014, 2011a), the functional networks were found without prior information/models. Functional connectivity was estimated by the Pearson correlation of the time courses of the independent components.

Significant connections during neurofeedback were found between IC9-IC22 and IC26-IC44. The brain regions corresponding to these ICs are presented in Table 1, consistent with the function and role of these regions reported during emotion regulation and retrieving autobiographical memories (Zotev et al., 2018, 2014, 2011a; Dehghani et al., 2019).

The 9th IC includes cuneus, precuneus, ventrolateral prefrontal cortex (VLPFC), dorsolateral prefrontal cortex (DLPFC) and posterior cingulate cortex (PCC). The 22nd IC contains amygdala, caudate, hippocampus, insula, putamen, and thalamus. The 26th IC contains cuneus, middle occipital, fusiform, and lingual gyrus. The 44th IC includes ventral striatum, ventrolateral prefrontal cortex (VLPFC), dorsolateral prefrontal cortex (DLPFC), orbitofrontal cortex (OFC), inferior parietal and middle temporal gyrus. Therefore, the main regions of IC9 are in the occipital, parietal, prefrontal cortex and limbic system and those of IC22 are in the limbic system and sub-lobar. Similarly, for IC26, they are in occipital and for IC44, they are in the limbic system, parietal, temporal, and prefrontal/frontal cortex regions. Functional connectivity analysis revealed significant correlation of IC9 and IC22 between the occipital/limbic/prefrontal cortex/parietal and limbic/sub-lobar networks and, for IC26 and IC44, between occipital and limbic/parietal/temporal/prefrontal (frontal) networks. Notably, as shown in Figs. 4 and 6, the functional connectivity between the two pairs of networks during upregulation is higher than the view blocks. This indicates additional coherence and synchronization among the brain regions during upregulation of positive emotion.

According to the emotion regulation model, emotion is generated by subcortical regions (amygdala and ventral/dorsal striatum) (Kohn et al., 2014). The trigger for emotion generation is by visual stimulus of individual positive images of autobiographical memories. The visual information of images is processed in the occipital lobe while the emotion is generated by the subcortical regions, especially the amygdala. Other processes and steps of emotional regulation, including the a) appraisal of stimulus, b) detection and initiation of regulation and finally, c) regulation and generation of new emotional states are performed by brain regions like VLPFC, DLPFC, MTG, insula, and amygdala (Kohn et al., 2014). Therefore, increased connectivity between occipital regions and frontal, limbic system (amygdala and ventral/dorsal striatum including putamen and caudate), and insula as a hub of receiving emotional signals from other brain regions in IC9-IC22 and IC26-IC44 is likely related to the processing of the visual stimulus by the occipital regions in the beginning of the three steps of emotion regulation (Chen et al., 2009; Kohn et al., 2014; Pohl et al., 2013). As mentioned by (Seeley et al., 2019), functional connectivity changes between insula and occipital region (cuneus) and between PFC and occipital regions (cuneus, lingual gyrus, middle occipital and fusiform) are significant in decreasing negative emotion. They seem to behave similarly in increasing positive emotion (based on neurofeedback and paradigm in this study) as reported in connections 9-22 and 26-44. The increased functional connectivity between cuneus and thalamus and between cuneus and hippocampus (IC9 and IC22) may be interpreted according to the role of hippocampus in memory retrieval that may initiate by images of positive autobiographical memories, and thalamus in relaying sensory information to occipital regions (Cerqueira et al., 2008; Jin and Maren, 2015; Wei et al., 2019).

The rise of functional connectivity between posterior cingulate cortex and hippocampus (IC9 and IC22) is because of their involvement in the memory system (Cheng et al., 2018).

The coactivation pattern between amygdala and VLPFC is according to the first and third steps of the emotion regulation model, where the appraisal of the stimulus occurs via a signaling from amygdala and basal ganglia to VLPFC through insula and generation of emotion response occurs through a feedforward signal from VLPFC to amygdala (Kohn et al., 2014). The rise of functional connectivity between insula and VLPFC/DLPFC is according to their same roles in three steps of emotion regulation model (Kohn et al., 2014). Increased connectivity between DLPFC and amygdala is related to the third step of the emotion regulation model and reciprocal anatomical connections in frontal brain regions and also between frontal and limbic system.

The increased connectivity between PCC and amygdala is mediated by the structural projections of the cingulate cortex to amygdala and the prefrontal cortex (Stevens et al., 2014). Decreased functional connectivity between PCC/precuneus and amygdala was associated with higher anxiety and the ability to regulate emotion and decrease in this symptom was associated with greater functional connectivity between PCC/precuneus and amygdala (Hahn et al., 2011;

Hamm et al., 2014; Li et al., 2016). Therefore, following emotion regulation, increased happiness, and decreased sadness (and also decreased anxiety according to psychometric tests), the functional connectivity between PCC and amygdala enhanced (IC9-IC22).

Parietal and temporal regions are involved in attention deployment and also integration of sensory information (Aday et al., 2017; Johnson, 2001; Sheldon et al., 2016). Therefore, the co-activation pattern between temporo/parietal regions and occipital regions (IC26-IC44) is potentially related to the processing of visual-sensory information and attention deployment.

Precuneus, as a region of parietal lobe, plays a key role in visual processing and visual attentional deployment, retrieving autobiographical memories, and direct regulation of amygdala activity to emotional stimulus through attentional deployment (Daselaar et al., 2008; Ferri et al., 2016). Previous studies (Hahn et al., 2011; Hamm et al., 2014; Li et al., 2016; Sato et al., 2019) revealed that increased functional connectivity between amygdala and precuneus was associated with subject happiness scale (and decreased anxiety), similar to neurofeedback paradigm used in our study. Therefore, according to the role of precuneus in attentional deployment to visual images of positive autobiographical memories and emotion generation by amygdala, the functional connectivity between amygdala and precuneus increased. Increased functional connectivity between amygdala and precuneus was also correlated with regulation success and symptom improvement, using positive autobiographical memories and neurofeedback in major depression disorder (Stevens et al., 2014; Young et al., 2018a, 2016). Increased functional connectivity between precuneus and hippocampus may be interpreted according to their same roles in retrieving autobiographical memory.

Overall, these findings provide significant correlation between brain regions from frontal, occipital, temporal and limbic system. They can be used as markers for treatment, treatment monitoring, or design of new treatment paradigms according to the interactions of the brain networks (connectivity based neurofeedback). Most of the identified networks (especially for regions located in prefrontal/frontal cortex) have a peak-value in the left hemisphere. According to the approach – withdrawal hypothesis, more activity of the left hemisphere is associated with emotions like joy (Davidson, 1998; Davidson et al., 1990), consistent with the neurofeedback paradigm and findings reported in this study. In the identified connections, the co-activation pattern between the prefrontal/frontal regions and other regions (limbic and occipital) provides valuable information for practical neuromodulation applications especially neurofeedback, where the prefrontal activity can be measured by EEG and enhancement/modulation of other brain regions can be predicted.

While the FCN results of this study are consistent with those of the previous studies and emotion regulation models, some of the connections identified in this study were not found in the previous studies. These connections are between a) occipital (fusiform, cuneus, middle occipital, and lingual gyrus) and other regions including limbic system/sub-lobar (thalamus, hippocampus, amygdala, caudate, putamen, insula, and ventral striatum), prefrontal/frontal cortex (DLPFC, VLPFC, and OFC), inferior parietal, and middle temporal gyrus and b) PCC and hippocampus. Table 2 compares brain functional connectivity found in previous studies with those of this study.

Simultaneous recording of EEG and fMRI for neurofeedback is expensive and also time consuming due to the practical problems such as installation of the EEG electrodes on the participant's head. Therefore, the sample size of our study (n=32) is relatively small but similar to or higher than those of the related previous studies (Perronnet et al., 2017; Zotev et al., 2014). Notably, among the 55 studies reviewed (Linhartová et al., 2019), 50 studies had sample sizes similar or less than ours. Moreover, each participant in this study had 10 runs of three blocks, namely, rest, view, and upregulation. The reliability of the ICA derived networks was additionally assessed using ICASSO (20 iterations). Due to previous reports of gender-specific differences in emotion regulation and process (Mak et al., 2009a, 2009b; McRae et al., 2008; Whittle et al., 2017), all participants in this study were male. Future studies may be conducted on the female subjects using the same emotion regulation paradigm.

5- Conclusion

The EEG neurofeedback along with induced positive autobiographical memories were used to modulate the activity and connectivity of brain regions during emotion regulation. Effectiveness of EEG neurofeedback in changing activity of brain regions during emotion regulation was demonstrated in previous studies. In contrast to the previous model based studies (Ferri et al., 2016; Li et al., 2016; Murphy et al., 2016; Seeley et al., 2019; Zotev et al., 2013, 2011a), group ICA was used as a global data-driven method to identify and characterize functional connectivity among all brain networks involved in neurofeedback. Experimental results revealed effects of neurofeedback in emotion regulation as increased interaction/connectivity among multiple brain regions, including prefrontal, limbic, temporal, and occipital regions. The brain connections identified in this work include those reported in the previous studies (Table 2) but also include additional connections between a) occipital (fusiform, cuneus, middle occipital, and lingual gyrus) and other regions including limbic system/sub-lobar (thalamus, hippocampus, amygdala, caudate, putamen, insula, and ventral striatum), prefrontal/frontal cortex (DLPFC, VLPFC, and OFC), inferior parietal, and middle temporal gyrus and b) PCC and hippocampus. Notably, the co-activation pattern between the prefrontal/frontal regions and other regions, obtained in this study, provides valuable information for practical neuromodulation applications especially EEG neurofeedback, where the prefrontal activity can be measured by EEG and enhancement/modulation of other brain regions can be predicted.

In majority of mental disorders, multiple brain regions and networks show abnormality. Novel neurofeedback methods based on the connectivity of the brain regions involved in emotion regulation may lead to increased treatment effectiveness. In particular, our connectivity analysis results may be used in future neurofeedback studies, instead of the traditional activity based neurofeedback. Modulation of the neural networks related to emotion regulation may provide new therapeutic paradigms for mental disorders such as MDD and PTSD to monitor both behavior and cognition, which are modulated by neurofeedback and other interventions (Zotev et al., 2011b).

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Author Contributions

All authors listed have made a substantial, direct and intellectual contribution to the work, and approved it for publication.

Conflict of interest

The authors have no conflict of interests or financial disclosures to declare.

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Figure Legends

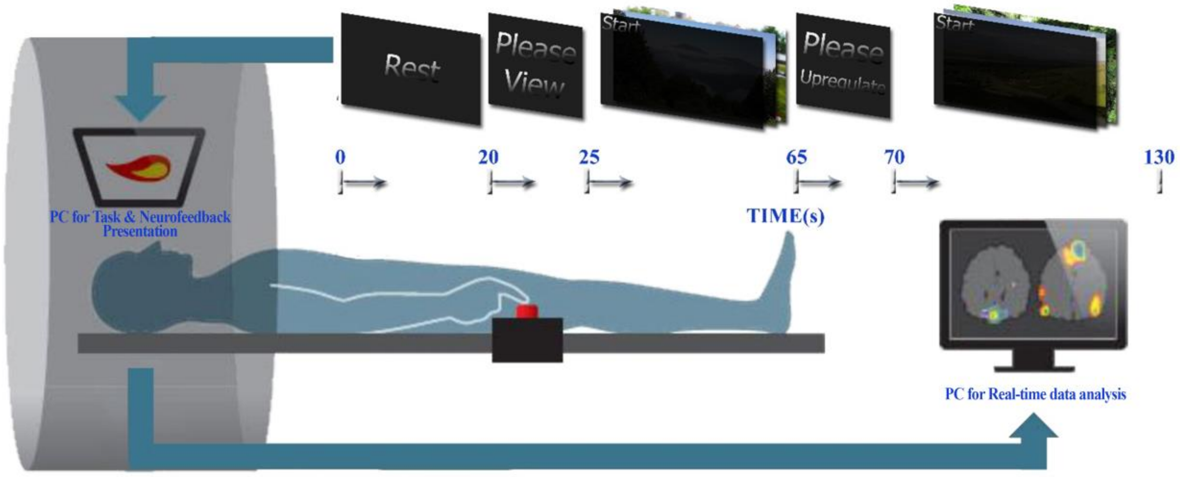


Fig. 1: The timing of rest, view, and upregulation blocks for each run of neurofeedback.

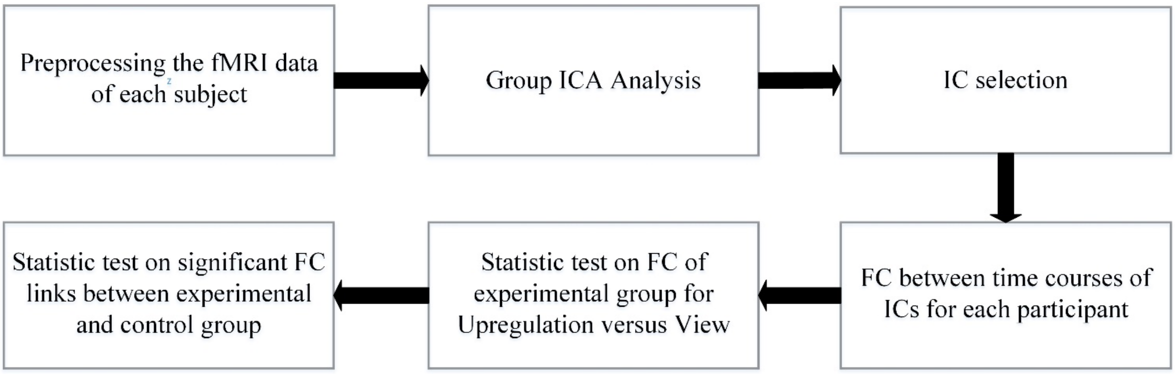
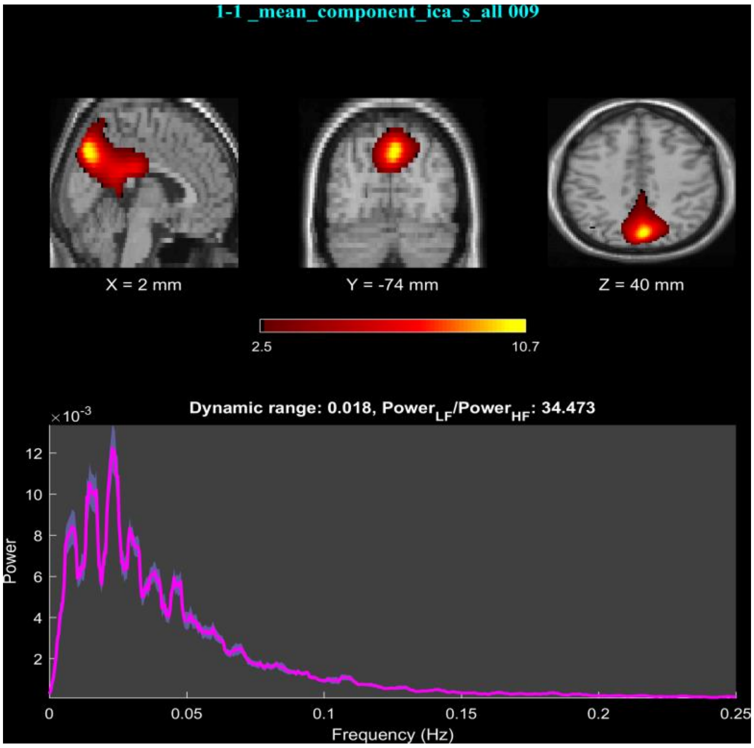
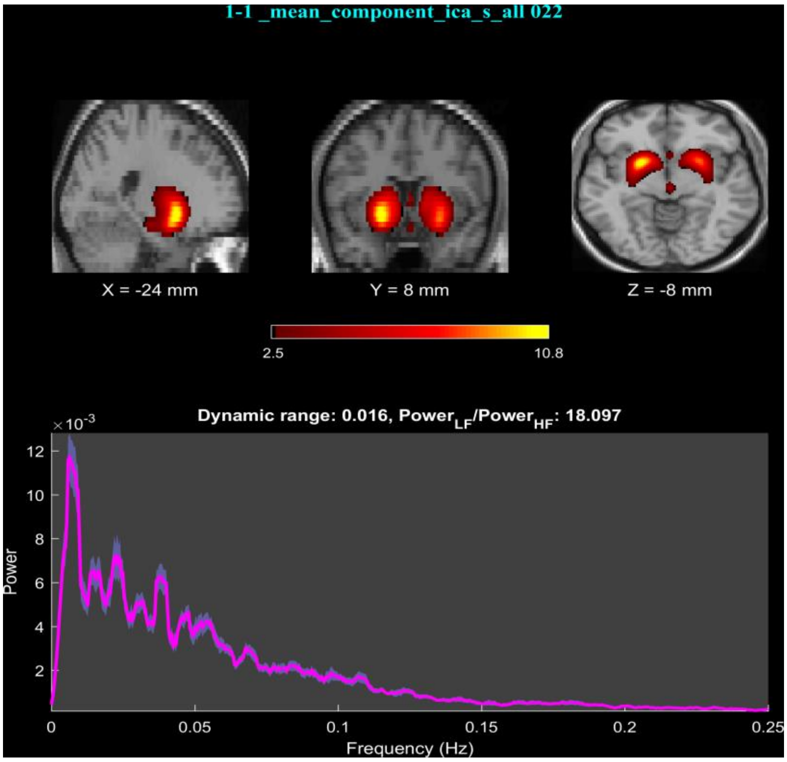
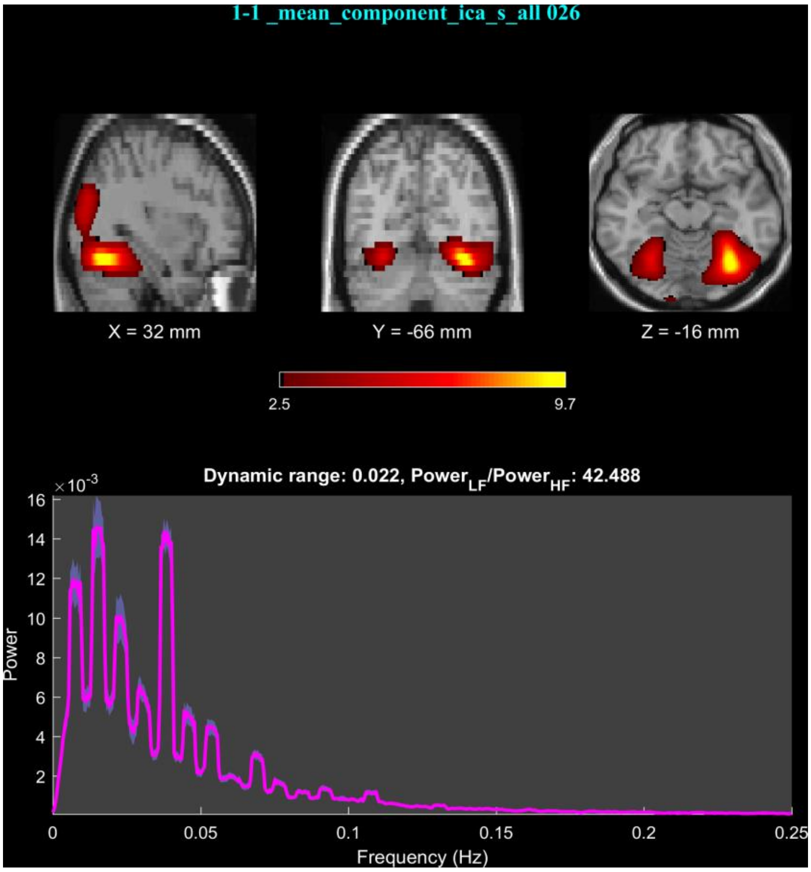


Fig. 2: Block diagram of offline analysis.



Brain Connectivity
Global Data-Driven Analysis of Brain Connectivity during Emotion Regulation by EEG Neurofeedback (DOI: 10.1089/brain.2019.0734)
This paper has been peer-reviewed and accepted for publication, but has yet to undergo copyediting and proof correction. The final published version may differ from this proof.


1-1 _mean_component_ica_s_all 022
X = -24 mm
Y = 8 mm
Z = -8 mm
2.5
10.8
Dynamic range: 0.016, Power$_{LF}$/Power$_{HF}$: 18.097
×10^{-3}
12
10
8
6
4
2
Power
0
0.05
0.1
0.15
0.2
0.25
Frequency (Hz)



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X = 32 mm
Y = -66 mm
Z = -16 mm
2.5
9.7
Dynamic range: 0.022, Power$_{LF}$/Power$_{HF}$: 42.488
$\times 10^{-3}$
16
14
12
10
8
6
4
2
Power
0
0.05
0.1
0.15
0.2
0.25
Frequency (Hz)

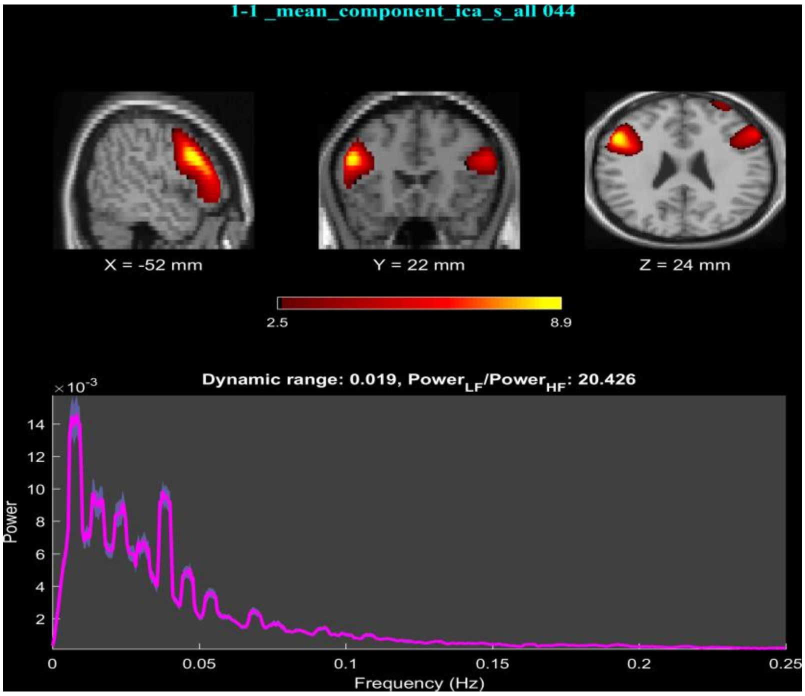


Fig. 3: Spatial maps and power spectrum of the time courses corresponding to components 9 (a), 22 (b), 26 (c), and 44 (d). The spatial maps are normalized to z-scores and thresholded at 2.5.

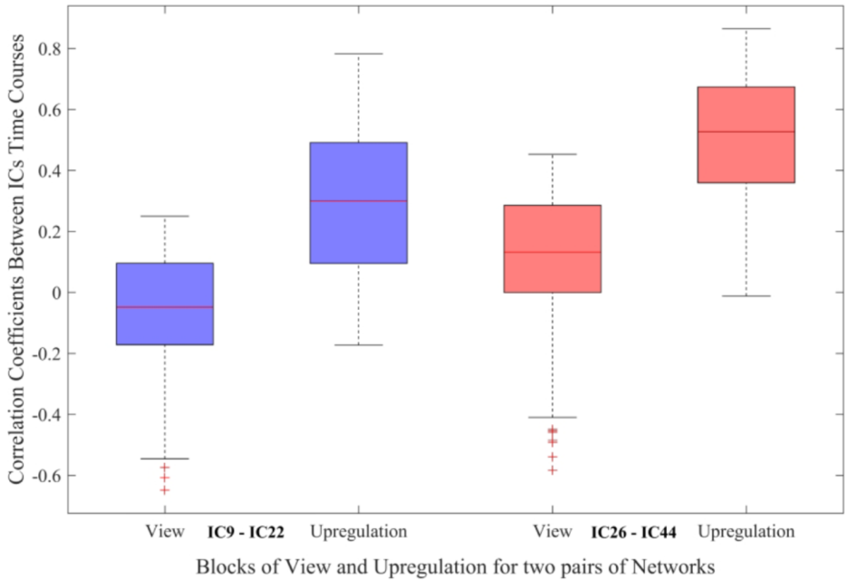


Fig. 4: Functional network connectivity between significant connections (FDR-corrected for multiple comparison at q=0.05) for the view and upregulation blocks in the experimental group.

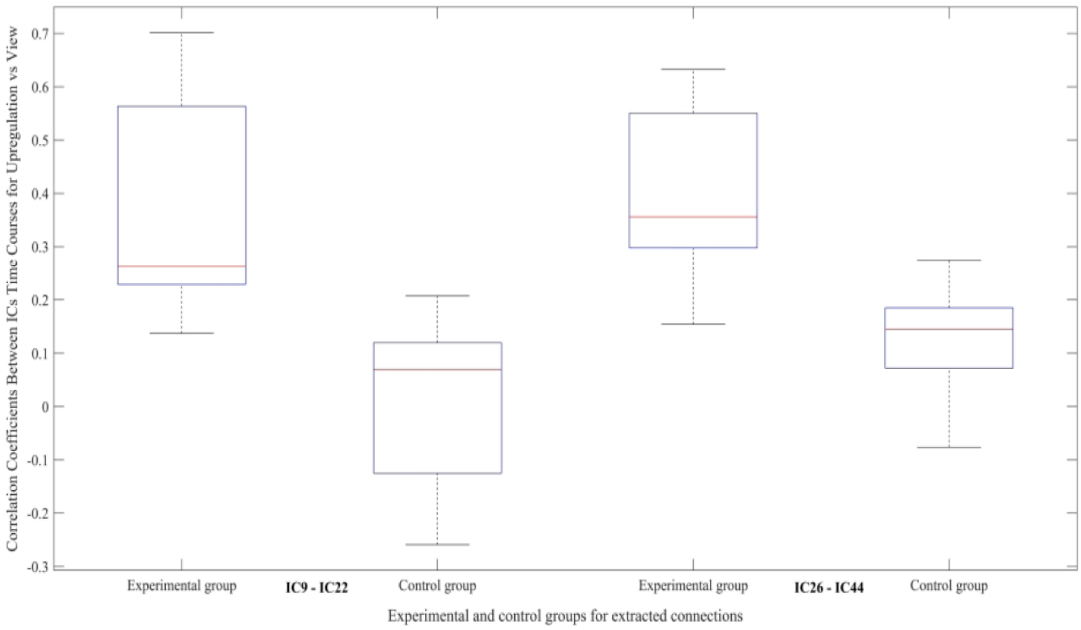


Fig. 5: Difference between functional connectivity of upregulation and view of ICs 9-22 and 26-44 in the experimental and control groups.

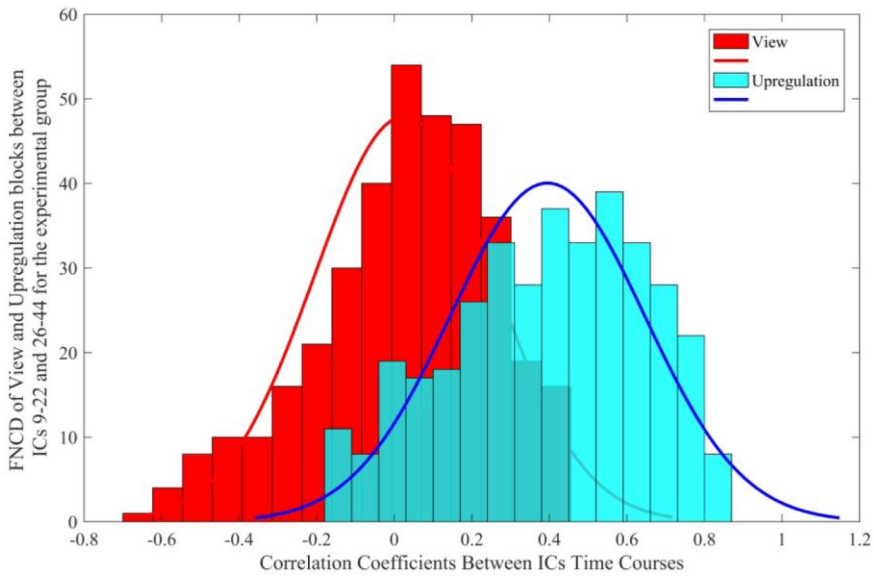


Fig. 6: Functional network connectivity distribution for the view and upregulation blocks of the two identified connections (between ICs 9-22 and 26-44) for the experimental group.

Table 1: Brain regions identified by components 9, 22, 26, and 44.

IC9	Max Z - MNI coordinate (x,y,z)
Left/Right Cuneus	10.2 (0,-72,34) / 9.1 (4,-74,34)
Left/Right Precuneus	10.8 (2,-74,38) / 10.8 (2,-74,40)
Left/Right Posterior Cingulate Cortex	7.5 (0,-42,22) / 7.1 (2,-42,22)
Left/Right Ventrolateral Prefrontal Cortex	2.4 (-45,16,4) / 2.2 (62,25,5)
Left/Right Dorsolateral Prefrontal Cortex	2.3 (-26,42,18) / 2.1 (42,-4,53)
IC22	Max Z - MNI coordinate (x,y,z)
Left/Right Amygdala	9.4 (-22,2,-16) / 8.4 (24,6,-16)
Left/Right Caudate	5.8 (-12,8,-12) / 5.9 (16,16,-6)
Left/Right Hippocampus	5.4 (-18,-4,-20) / 4.7 (22,-2,-20)
Left/Right Insula	8.5 (-26,8,-14) / 5.5 (28,10,-16)
Left/Right Putamen	10.8 (-24,8,-8) / 8.7 (24,8,-10)
Left/Right Thalamus	2.9 (-12,-6, 6) / 2.6 (4,-8,4)
IC26	Max Z - MNI coordinate (x,y,z)
Lef Cuneus	3.4 (-6,-98, 6)
Left/Right Middle Occipital	3.9 (-14,-104,-8) / 5.5 (36,-84,8)
Left/Right Fusiform	6.3 (-30,-62,-16) / 9.9 (32,-66,-18)
Left/Right Lingual Gyrus	4.7 (-26,-66,-12) / 6.6 (22,-68,-14)
IC44	Max Z - MNI coordinate (x,y,z)
Left/Right Ventral Striatum	8.1 (-14,22,-3) / 5.1 (12,18,-6)
Left/Right Ventrolateral Prefrontal Cortex	8.9 (-52,22,24) / 6.3 (54,26,22)
Left/Right Dorsolateral Prefrontal Cortex	6.6 (-52,14,36) / 4.7 (52,34,20)
Left Orbitofrontal Cortex	5.1 (-50,42,-4)
Right Inferior Parietal	3.2 (60,-48,38)
Right Middle Temporal Gyrus	2.6 (64,-18,-20)

Table 2: Comparison of brain connectivity changes in emotion regulation studies and this study (as a result of neurofeedback).

Study #	References	Brain area
1	(Young et al., 2018)	amygdala and: inferior frontal G/lateral, medial PFC, medial frontopolar Cortex, ventrolateral PFC, medial frontal Gyrus, ACC, insula, ventral striatum, putamen, thalamus, precuneus, cerebellum, temporal pole
2	(Paret et al., 2016b)	amygdala and: supplementary motor area, middle frontal gurus, brain stem, precuneus, ventromedial prefrontal cortex, white matter/ right putamen/insula
3	(Zotev et al., 2011b)	amygdala and : frontal lobe (Ventrolateral prefrontal cortex, orbitofrontal cortex, Middle frontal gyrus, superior frontal gyrus, ventromedial prefrontal cortex), temporal Lobe (middle temporal gyrus), limbic Lobe (hippocampus), sub-lobar Regions (insula, thalamus)
4	(Paret et al., 2016a)	amygdala and: DLPFC, Precentral gyrus, paracentral lobe, parahippocampal gyrus, extending to thalamus and hippocampus, cerebellum
5	(Herwig et al., 2019)	amygdala and: ACC, DLPFC, DMPFC, pre-SMA and VLPFC
6	(Koush et al., 2017)	amygdala-DLPFC
7	(Nicholson et al., 2017)	amygdala and: DMPFC, DLPFC, ACC
8	(Sarkheil et al., 2015)	PFC-PCC
9	(Veit et al., 2012)	insula and: lingual gyrus, ventrolateral PFC, Frontal inferior operculum, inferior orbitofrontal, Middle frontal, Middle orbitofrontal, Occipital inferior, Dorsal medial PFC
10	(Cohen Kadosh et	insula and: mid cingulate cortex, supplementary motor area,

	al., 2016)	amygdala
11	(Ros et al., 2013)	ACC and mid-cingulate cortex.
12	(Banks et al., 2007)	amygdala and: OFC and DMPFC
13	(Morawetz et al., 2017)	inferior frontal gyrus (IFG) and: DLPFC, DMPFC, VMPFC, ACC, amygdala amygdala and prefrontal regions (e.g. DMPFC)
14	(HE et al., 2017)	amygdala and: ACC amd thalamus PCC and centromedial amygdala (subregions of amygdala)
15	(Koush et al., 2019)	temporoparietal junction (TPJ) and: SFG, DMPFC, vmPFC vmPFC and: DMPFC, amygdala,
16	This study	Occipital regions (cuneus, fusiform, lingual gyrus, middle occipital) and: Limbic system and sub-lobar (thalamus, hippocampus, amygdala, caudate, putamen, insula, ventral striatum), prefrontal and frontal cortex (VLPFC, DLPFC and OFC), parietal (inferior parital), temporal (middle temporal gyrus) Parietal (precuneus) and: amygdala, hippocampus PCC and: amygdala, hippocampus VLPFC/DLPFC and: amygdala, insula

1- Psychometric Testing

To measure positive and negative mood state induced as a result of neurofeedback, each participant completed a short Persian version of the Profile of Mood States (POMS) (Spielberger, 1972) and the complete Persian version of the positive-negative affect scale (PANAS) (Watson et al., 1988) before and after neurofeedback. For the experimental group, the average scores for the PANAS did not change significantly from before to after neurofeedback but for the PANAS positive and negative mood states and POMS, the changes were significant (paired t-test; df=17; $p_{\text{positive mood states of PANAS}} = 4.5 \times 10-4, \text{pnegative mood states of PANAS} = 2 \times 10-3, \text{pPOMS} = 1.1 \times 10^{-3}$).

For the experimental group, the mean ± standard deviation of POMS for "before neurofeedback" was 24.6 ± 10.9 and for "after neurofeedback" was 17 ± 6.9, of PANAS for "before neurofeedback" was 52.2 ± 11.5 and for "after neurofeedback" was 51.6 ± 8.8, of PANAS positive mood state for "before neurofeedback" was 31.4 ± 6.1 and for "after neurofeedback" was 37.5 ± 6.4, of PANAS negative mood state for "before neurofeedback" was 20.8 ± 7.2 and for "after neurofeedback" was 14.1 ± 4.8. The results demonstrated that using neurofeedback and retrieving positive autobiographical memories, the participants could increase positive emotion and decrease negative emotion.

For the control group, the result of none of the psychometric tests changed significantly from before to after neurofeedback (paired t-test; df=13; $p_{\text{PANAS}} = 0.1$, $p_{\text{positive mood states of PANAS}} = 0.33$, $p_{\text{negative mood states of PANAS}} = 0.64$, $p_{\text{POMS}} = 0.08$). The mean ± standard of POMS for "before neurofeedback" was 27.1 ± 11.3 and for "after neurofeedback" was 22.8 ± 11.1, of PANAS for "before neurofeedback" was 54.2 ± 5.9 and for "after neurofeedback" was 53 ± 4.8, of PANAS positive mood state for "before neurofeedback" was 32.1 ± 5.7 and for "after neurofeedback" was 33.7 ± 5.6, of PANAS negative mood state for "before neurofeedback" was 22.1 ± 6 and for "after neurofeedback" was 19.3 ± 5. The control group showed increased positive emotion and decreased negative emotion using recalling autobiographical memory even with sham neurofeedback (as a result of retrieving positive autobiographical memories).

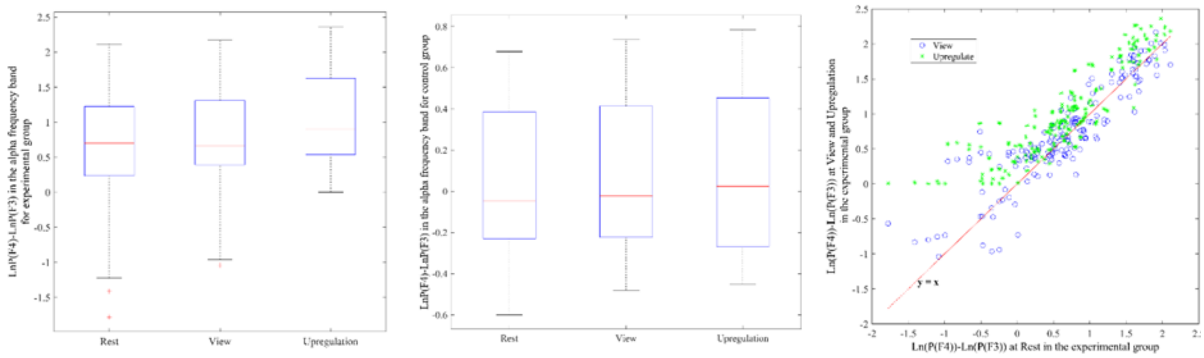
To measure the anxiety induced by fMRI scanning, the Persian version of Beck Anxiety Inventory (BAI) (Ghassemzadeh et al., 2005) and State Anxiety Inventory of STAI (Azizi et al., 2013) were completed by each participant before and after the neurofeedback. The mean ± standard deviation of the Beck Anxiety Inventory (BAI) and State Anxiety Inventory of all participants for "before neurofeedback" were 8 ± 4.4 and 27 ± 2.5 and for "after neurofeedback" was 6.6 ± 4.7 and 26.3 ± 2.8. Therefore, the level of anxiety decreased after the experiment as a result of emotion regulation and neurofeedback.

Samples of the positive autobiographical memories written by the subjects are as following:

- Winning the match/game
- Wedding or marriage ceremony
- Admire, support, and encourage by others
- Party
- Meeting (friends, family, etc.)
- Trip to …
- Working at …
- First day of employment
- Birth of son/daughter, nephew,
- Holiday in …
- Memories with the first friend
- Going to cinema or movie with friends
- Watching the football match between X and Y in the stadium
- First day of university
- Admitted to university
- Concert
- Exercise
- Visiting a special place
- Barbecue with family/friends
- Helping ….
- Listening to music
- Driving a car in the Chaloos road
- Buying the first car/house

2- Neurofeedback Results

The effectiveness of the EEG frontal asymmetry was demonstrated in the previous studies for emotion regulation and treatment of mental disorders (Allen and Reznik, 2015; Mennella et al., 2017; Meyer et al., 2018; Quaedflieg et al., 2015). The effectiveness of the neurofeedback through retrieving positive autobiographical memories for this study was described in (Dehghani et al., 2019). The results of the experimental group showed larger frontal asymmetry changes during the upregulation blocks relative to the view and rest blocks (paired t-test; df = 179; $\text{p-value}_{\text{Upregulation-View}} = 5.1 \times 10^{-5}$, $\text{p-value}_{\text{Upregulation-Rest}} = 3.8 \times 10^{-6}$). Also, the changes of the frontal asymmetry between the upregulation and view/rest blocks of the experimental group were significantly different from those of the control group (two sample t-test; df = 318; $\text{p-value}_{\text{Upregulation-View (Experimental group-Control group)}} = 1.7 \times 10^{-4}$, $\text{p-value}_{\text{Upregulation-Rest (Experimental group-Control group)}} = 3.9 \times 10^{-3}$). The EEG frontal asymmetry changes in the alpha frequency band for Upregulation, View, and Rest in the experimental and control group are presented in Figures 1(a) and (b) (Dehghani et al., 2019). Figure 1(c) illustrated the EEG frontal asymmetry (LnP(F4)-LnP(F3)) of Upregulation and View versus Rest blocks in the experimental group. Since the feedback for the experimental group is based on the powers of channels F4 and F3, and the subjects try to increase LnP(F4)-LnP(F3) in the alpha frequency band versus the view period, we expect the mean power in the Upregulation blocks to be higher than the rest and view blocks as illustrated in Figure 1(a). In Figure 1(c) and based on the line y = x, the frontal asymmetry in the alpha frequency band for the most Upregulation blocks is higher than View and Rest blocks. The result demonstrates the successfulness of neurofeedback in the experimental group.



(a) (b) (c)

Figure 1: EEG frontal asymmetry in alpha frequency band for Rest, View and Upregulation in the (a) experimental (b) control group (c) scatter plot of frontal asymmetry in blocks of View and Upregulation versus Rest for the experimental group

According to the previous studies, the EEG frontal asymmetry can be used as a marker of emotional states and well-being (Cavazza et al., 2014). Based on the approach and withdrawal theory, increased frontal asymmetry is associated with the emotional states like happiness (Davidson et al., 1990).

A group analysis of the fMRI data revealed several activated brain regions from the limbic system, parietal, occipital, and the prefrontal regions with significant signal changes and high effect size (paired t-test for every one of the 118 ROIs; df = 179; FDR-corrected for multiple comparison; q = 0.01) during the upregulation blocks relative to the view and rest blocks in the experimental group with no significant changes for the control group. The activated brain regions were left/right amygdala, thalamus, insula, dorsomedial prefrontal cortex, caudate, cuneus, hippocampus, posterior cingulate cortex, orbitofrontal cortex, middle temporal gyrus, lingual gyrus, ventral striatum, dorsolateral prefrontal cortex, ventrolateral prefrontal cortex, superior parietal, inferior parietal, supramarginal, postcentral and anterior cingulate cortex. The results were consistent with the previous studies of emotion regulation and retrieving positive autobiographical memories (Bado et al., 2014; Johnston et al., 2011; Young et al., 2018; Zotev et al., 2018, 2014).

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